

The second exam is on **Wednesday, October 20, 2021**. Please bring pencil (0.7 mm-0.9mm), non-programmable calculator (**NCEES**), and eraser with you. Exam is closed book and closed notes. You must show the full details of your work for full credit. **No credit just for answer!** Storing the formulas or notes related to EMCH 290 is absolutely prohibited in the calculator. Please turn off your cell phone and laptop and keep them in your backpack. It is the responsibility of every student at the University of South Carolina-Columbia to adhere steadfastly to truthfulness and to avoid dishonesty, fraud, or deceit of any type in connection with any academic program. Any student who violates this Honor Code or who knowingly assists another to violate this Honor Code shall be subject to discipline.

You have **50 minutes** to finish your Exam (**9:40 AM – 10:30 AM**). Make-up exam will not be given unless by the approval of Mechanical Engineering Department. Please make sure to turn in your exam to me at the end of exam period regardless how you did! Missing exam will be considered **zero**. **If you arrive late, you will NOT be given extra time to complete the test.** **The students with Disability should reserve a seat a week before the date of exam at Disability Office with extended time.**

The exam # 2 will cover Evaluating Properties Chapter as it was discussed previously. Please make sure that you will understand the following materials

- a. Dimensions and Units (SI Units, English Engineering System)
- b. Energy Concepts, Energy Balance Equation for closed system (First Law of Thermodynamics).
- c. Points and Regions on the Property Diagram of a Pure Substance
- d. Phase change and $p - v$, $T - v$ and $P - T$ diagrams.
- e. Two-Phase Liquid –Vapor Mixture and quality (x) equations.
- f. Heating at constant pressure problems.
- g. Heating at constant volume problems.
- h. Stirring at constant volume problems.
- i. Analyzing 2 processes in series.
- j. Universal gas constant and equations of state.
- k. Generalized Compressibility Data or Z Chart.
- l. Analyzing an ideal gas undergoing a thermodynamic cycle.
- m. Using the energy balance and constant specific heats.

*The Exam is reasonable and easy if you understand the covered materials in the class, examples, and lectures.

**All Tables and Required Equations will be provided during exam.